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Aim: Simulation of a Basic Communication System (Transmitter Channel – Receiver) using MATLAB

Objective :

1. **To understand** the basic components of a communication system: Transmitter (Tx), Communication Channel, and Receiver (Rx).
2. **To generate** a message signal and a carrier signal using MATLAB.
3. **To perform** Amplitude Modulation (AM) for transmitting the message signal.
4. **To simulate** the effect of noise in the communication channel and observe its impact on the transmitted signal.
5. **To recover and analyze** the original message signal at the receiver using demodulation.

Theory :

Theory of Basic Communication System :

A **communication system** is a system that transfers information from one place to another accurately and efficiently. The information may be in the form of **voice, text, images, video, or sensor data**. The primary goal of any communication system is to ensure that the information sent by the transmitter reaches the receiver with minimum distortion and maximum reliability.

Communication systems are classified into two types:

- **Analog Communication System:** Transmits continuous-time signals such as voice and audio. Examples include AM and FM radio.
- **Digital Communication System:** Transmits digital (binary) data in the form of 0s and 1s. Examples include mobile communication, Wi-Fi, and the Internet.

A basic communication system consists of three major blocks:

1. Information Source

The information source generates the message that needs to be transmitted. The message may be:

- Voice from a microphone
- Music from an audio player
- Video from a camera
- Data from a computer
- Sensor readings in IoT devices

This information is usually called the **baseband signal** or **message signal**.

2. Transmitter (Tx)

The transmitter processes the message signal and prepares it for transmission through the communication channel.

The functions of the transmitter include:

a) Signal Generation

The message signal is generated from the source.

Example:

- Voice signal
- Audio signal
- Digital data

b) Modulation

The message signal is combined with a high-frequency carrier signal.

Modulation is necessary because:

- Low-frequency signals cannot travel long distances efficiently.
- Antenna size becomes practical.
- Multiple signals can be transmitted simultaneously.
- Noise effects are reduced.

Common modulation techniques include:

- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)
- ASK
- FSK
- PSK
- QPSK
- QAM

In this experiment, **Amplitude Modulation (AM)** is used.

The mathematical expression of an AM signal is

$s(t)=Ac[1+kam(t)]\cos \quad (2\pi fct)s(t)=A_c[1+k_a\, m(t)]\cos(2\pi\, f_c\, t)s(t)=Ac[1+ka\, m(t)]\cos(2\pi fct)$

where

- AcA_cAc = Carrier amplitude
- m(t)m(t)m(t) = Message signal
- kak_aka = Modulation index
- fcf_cfc = Carrier frequency

c) Amplification

After modulation, the signal is amplified to provide sufficient power for transmission.

d) Transmission

The amplified signal is sent through an antenna or transmission medium.

3. Communication Channel

The communication channel is the medium through which the signal travels from the transmitter to the receiver.

The channel can be:

Guided (Wired)

- Twisted pair cable
- Coaxial cable
- Optical fiber

Unguided (Wireless)

- Radio waves
- Microwave
- Satellite
- Infrared

Effects of Communication Channel

During transmission, the signal may experience:

Noise
Random unwanted electrical signals that distort the original message.
Examples:
- Thermal noise
- Atmospheric noise
- Industrial noise

In MATLAB, noise is usually simulated using **Additive White Gaussian Noise (AWGN).**

Interference
Signals from other communication systems mix with the desired signal.
Examples:
- Mobile phone interference
- Radio interference

Attenuation
Reduction in signal strength as it travels through the medium.

Distortion
Change in the waveform shape during transmission.

Fading
Variation in received signal strength due to multiple propagation paths.

4. Receiver (Rx)
The receiver extracts the original information from the received signal.
The receiver performs the following operations:

a) Reception
The antenna receives the transmitted signal.

b) Filtering
Removes unwanted high-frequency noise.
c) Demodulation
Recovers the original message from the modulated signal.
For AM systems, an **Envelope Detector** is commonly used.
d) Amplification
The recovered signal is amplified if necessary.
e) Output
The original information is delivered to the user.

Examples:

- Speaker
- Display
- Computer
- Mobile phone

Working of the Communication System

The communication process follows these steps:

1. The information source generates the message signal.
2. The transmitter modulates the message using a carrier signal.
3. The modulated signal is transmitted through the communication channel.
4. Noise and interference may affect the transmitted signal.
5. The receiver receives the noisy signal.
6. Demodulation is performed to recover the original message.
7. The recovered message is delivered to the destination.

Communication System Flow



Need for Modulation

Modulation is an essential process in communication systems because it:

- Enables long-distance transmission.
- Reduces the size of transmitting and receiving antennas.
- Prevents mixing of different signals.
- Allows frequency multiplexing.
- Improves transmission efficiency.
- Enhances immunity to noise.

Without modulation, transmitting low-frequency signals directly would require impractically large antennas and would be more susceptible to interference.

Advantages of a Communication System

- Enables long-distance communication.
- Fast transfer of information.
- Supports wireless communication.
- Reliable transmission of data.
- Easy integration with modern digital systems.
- Foundation for Internet, mobile communication, and satellite systems.

Applications of Communication Systems

- AM/FM Radio Broadcasting
- Television Broadcasting
- Mobile Communication (2G/3G/4G/5G)
- Wi-Fi Networks
- Bluetooth Communication
- Satellite Communication
- Optical Fiber Communication
- Internet and Computer Networks
- Military Communication Systems
- Medical Telemetry and Remote Patient Monitoring
- IoT (Internet of Things) Devices
- GPS Navigation Systems

Term	Description
Message Signal	Original information to be transmitted
Carrier Signal	High-frequency signal used to carry the message
Modulation	Process of combining the message with the carrier
Communication Channel	Medium through which the signal travels
Noise	Unwanted signal added during transmission
Demodulation	Process of recovering the original message
Transmitter	Device that prepares and sends the signal
Receiver	Device that receives and recovers the message

Observations

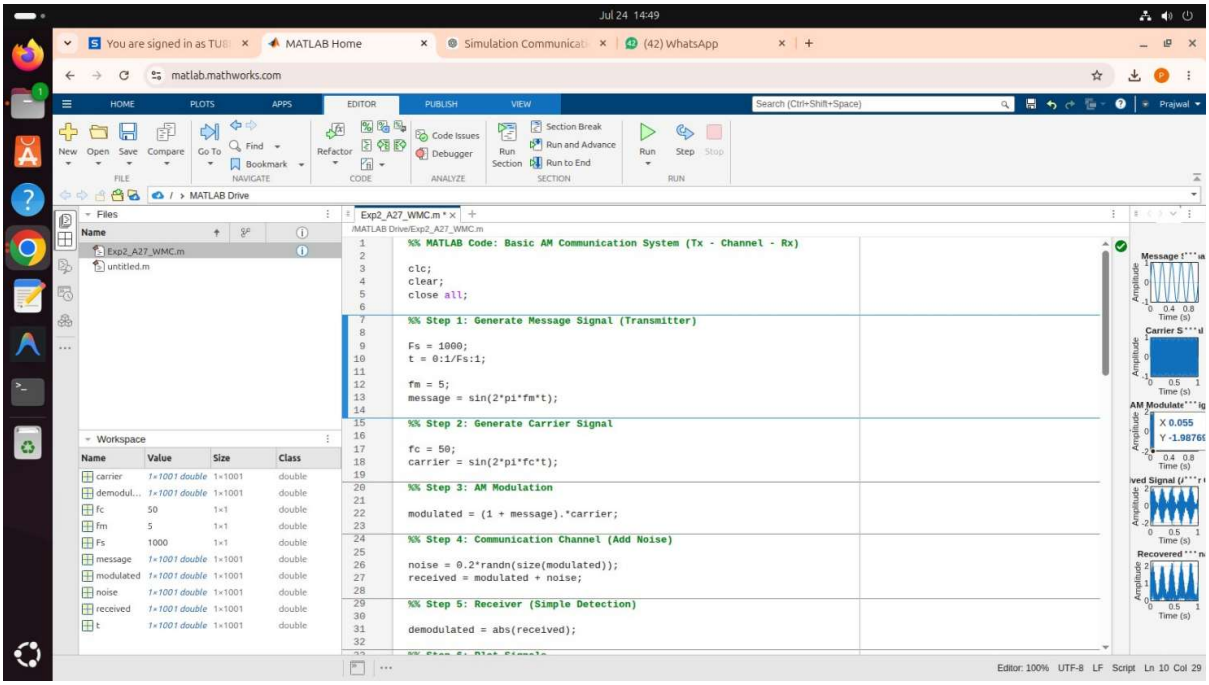
- The message signal is a low-frequency signal carrying the information.
- The carrier signal is a high-frequency signal used to transmit the message.
- The transmitted signal is formed by combining the message and carrier using AM.
- Noise added in the communication channel distorts the transmitted signal.
- The receiver demodulates the received signal to recover the original message.

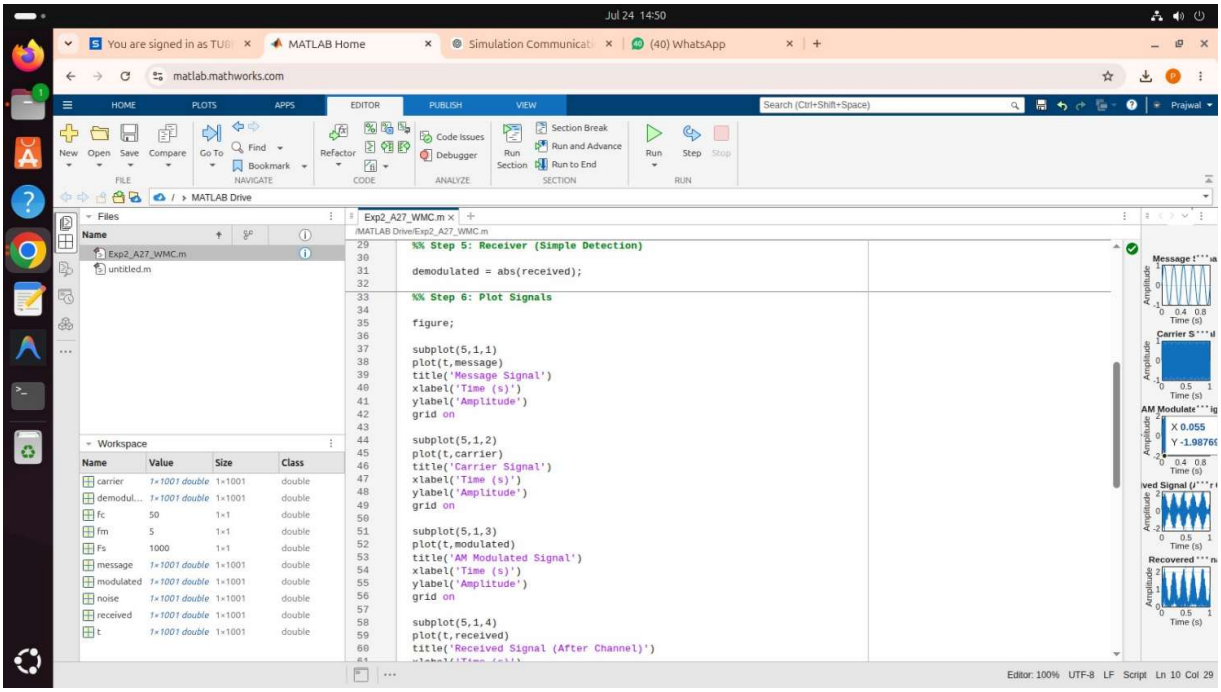
Parameter	Value
Message Frequency	5 Hz
Carrier Frequency	50 Hz
Sampling Frequency	1000 Hz
Modulation Technique	Amplitude Modulation (AM)
Channel	Additive Noise
Receiver	Simple Envelope Detector

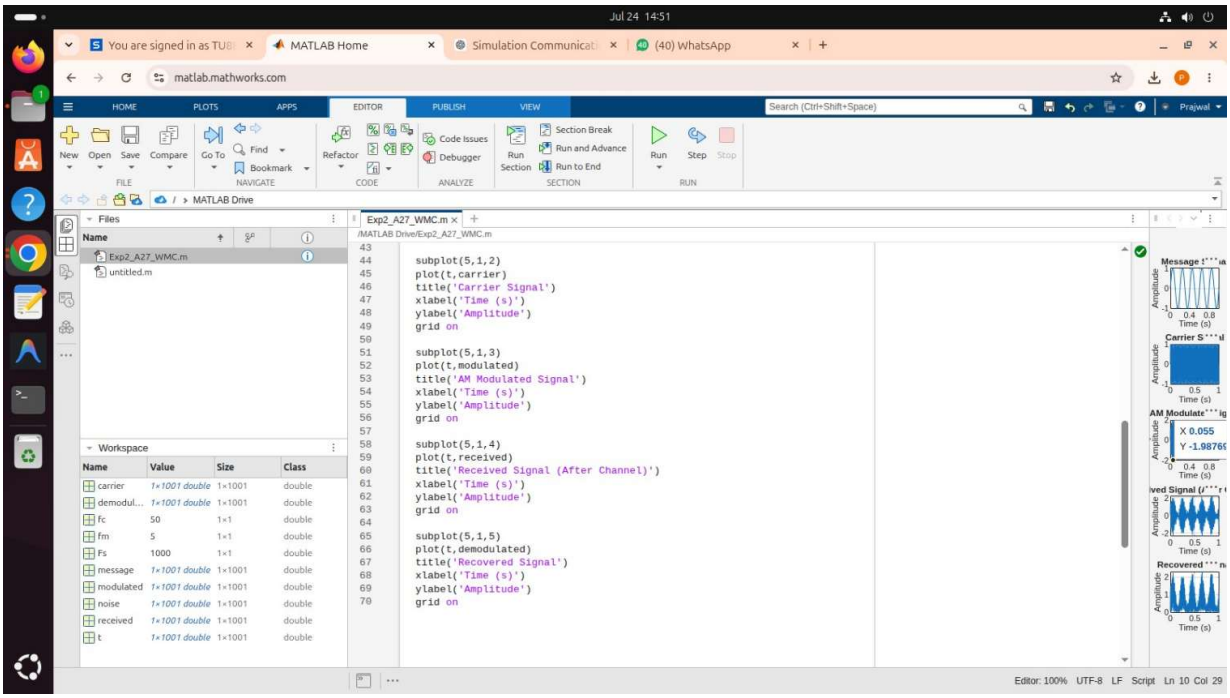
Conclusion :

The MATLAB simulation of the basic communication system was successfully carried out. The experiment demonstrated the complete communication process, including message signal generation, amplitude modulation (AM), transmission through a noisy channel, and demodulation at the receiver. It was observed that noise introduced in the communication channel affected the transmitted signal, but the receiver was able to recover the original message with acceptable accuracy. This experiment provided a clear understanding of the roles of the transmitter, communication channel, and receiver, as well as the importance of modulation and demodulation in achieving reliable and efficient signal transmission in modern communication systems.

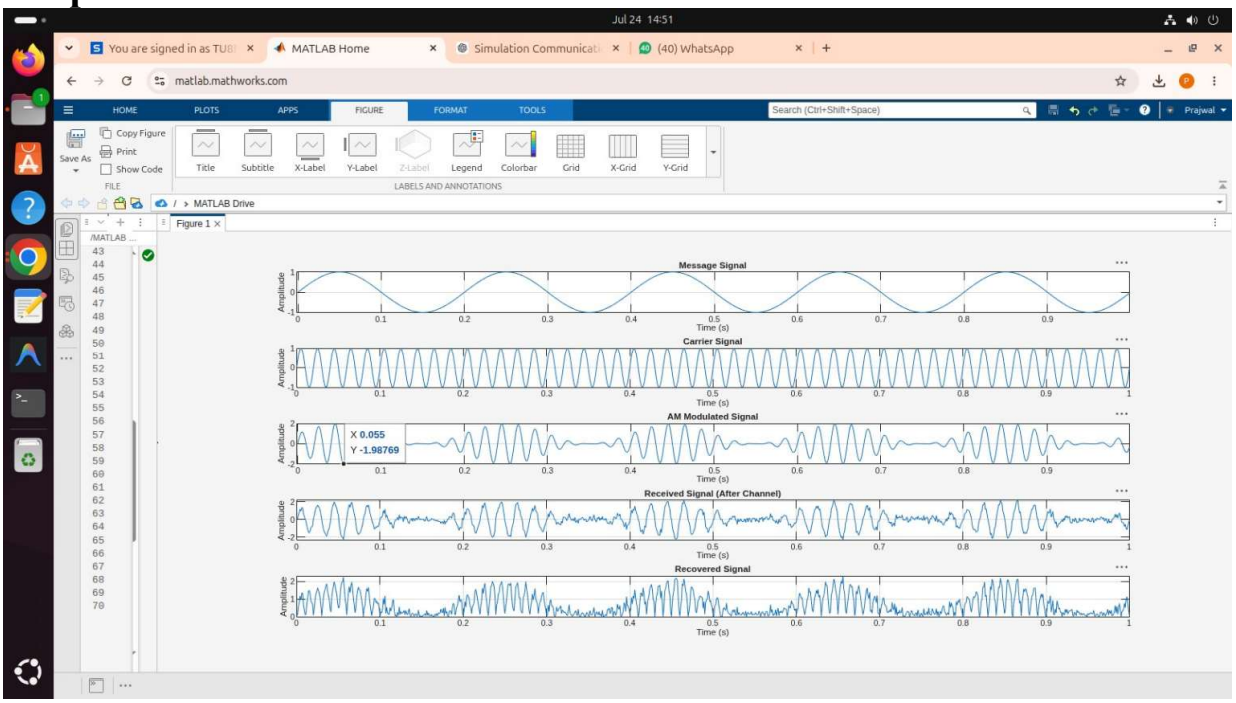
Code:




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Files
Exp2_A27_WMC.m
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Workspace
%% Step 5: Receiver (Simple Detection)
demodulated = abs(received);
%% Step 6: Plot Signals
figure;
subplot(5,1,1)
plot(t,message)
title('Message Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
subplot(5,1,2)
plot(t,carrier)
title('Carrier Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
subplot(5,1,3)
plot(t,modulated)
title('AM Modulated Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
subplot(5,1,4)
plot(t,received)
title('Received Signal (After Channel)')
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untitled.m
Workspace
subplot(5,1,2)
plot(t,carrier)
title('Carrier Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
subplot(5,1,3)
plot(t,modulated)
title('AM Modulated Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
subplot(5,1,4)
plot(t,received)
title('Received Signal (After Channel)')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
subplot(5,1,5)
plot(t,demodulated)
title('Recovered Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
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Output:


Message Signal
Carrier Signal
AM Modulated Signal
X 0.055
Y -1.98769
Received Signal (After Channel)
Recovered Signal
Amplitude
Time (s)